



Figure 6: Designer screen



Figure 7: Components view

Table 1

| Component name         | Purpose                          | Location                                  |
|------------------------|----------------------------------|-------------------------------------------|
| Label                  | To display a label               | Palette-->User Interface-->Label          |
| Button                 | To trigger events                | Palette-->User Interface-->Button         |
| Horizontal arrangement | To arrange the child components  | Palette-->Layout-->Horizontal Arrangement |
| Notifier               | To display on-screen information | Palette-->User Interface-->Notifier       |
| Firebase DB            | To store data persistently       | Palette-->Experimental-->Firebase DB      |



Figure 8: URL of the project

display some headings or markings on the screen.

2. **Button:** Buttons let you trigger the event and are very essential components.
3. **Horizontal arrangement:** This is a special component that keeps all child components aligned with it.
4. **Notifier:** This is used to display some instructions or give controls over your existing components. You will see its functionality in more detail as we implement it in our project.
5. **Firebase DB:** Fire DB, as you already know, is the cloud based database utility from Google. We will use it to store the user's data over the cloud.

Listed in Table 1 are the components that we will need for this application. We will drag them on to the designer from the left hand side palette.

1. Drag and drop the components mentioned in the table to the viewer.
2. Visible components will be visible to you, while the non-visible components will be located beneath the viewer under the tag *Non-visible*.
3. Labels, along with the respective text boxes, need to be put within the *Horizontal arrangement* so as to keep them aligned horizontally.
4. If you have dragged and placed everything, the layout will look something like what's shown in Figure 6.

5. Make the necessary property changes like we did in changing the text property for the label and button components.
6. Renaming the components helps to identify them in the block editor.
7. Your graphical user interface is now ready. Figure 6 shows how the application will look after the installation.
8. Figure 7 shows the hierarchy of the components that we have dragged to the designer.

If you are a bit confused after seeing the designer and the components viewer, let me explain things a bit. Here is the hierarchy for our application.

1. We have just placed four text boxes to get the details from the user.
  2. For easy reading, we have inserted a label before the text box.
  3. We need to configure the Firebase component to connect to the demo project we have created using the steps mentioned earlier. For that, just assign the URL of your project to the Firebase URL property of the Firebase component in the designer.
- Now, let's head towards the blocks editor to define the behaviours. Let's first discuss the actual functionality that we expect from our application.

1. The user should be able to write the text in the respective boxes.
2. Upon clicking the *Sign Up* button, all details should be saved to the database and a unique